

Technical Report No. 68.435.18.0510.01
Dated 2018-09-12

Client: TMA Germany GmbH

Address: Oberanger 6 80331 München

Attn.: Benedikt Klarmann

Sample Description: Glass cosmetic packaging with plastic closures

Model No.: J574/ J575/J400

Supplier: Guangzhou Jiaxing Glass Products Co., Ltd

Buyer: TMA Germany GmbH

Country of origin: China, Mainland

Exported to: Germany

Sample Received Date: 2018-08-23, 2018-09-11

Test Period: From 2018-08-23 to 2018-09-11

Location of Testing: TÜV SÜD Certification and Testing (China) Co., Ltd.
Shenzhen Branch

Purpose of examination: As specified by client, to test as regulated by the German Food & Feed Acts LFGB (§ 30 & 31) and Regulation (EC) No.1935/2004

Test Result: Refer to following page(s)

Remark: The result relates only to the items tested.

TÜV SÜD Certification and Testing (China) Co., Ltd. Shenzhen Branch
TÜV SÜD Group

Prepared by:



Jason Peng
Project Handler



Reviewed by:



Scarlett Liang
Designated Reviewer

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SUMMARY OF TEST RESULTS

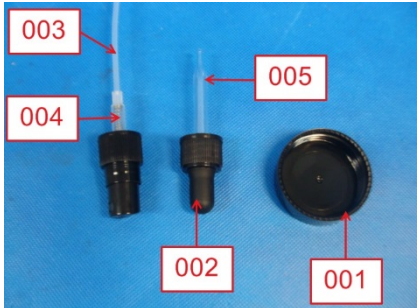
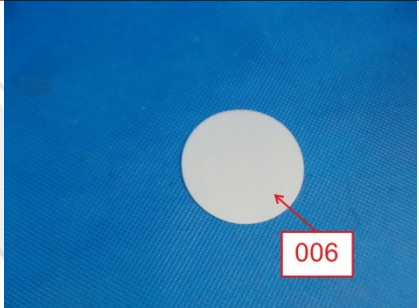
No.	Test Requested	Conclusion	Remarks
1.	For material: Plastics or coating – Overall migration test for compliance with regulation (EU) 10/2011 and it's amendment (EU) 2016/1416	PASS	
2.	For material: Plastics or coating – Specific Migration of Heavy Metals (Ba, Co, Cu, Fe, Li, Mn, Zn, Al, Ni) for compliance with regulation (EU) 10/2011 and it's amendments regulation (EU) 2016/1416, (EU) 2017/752	PASS	
3.	For material: Plastics or coating – Specific Migration of Primary Aromatic Amine for compliance with regulation (EU) 10/2011.	PASS	
4.	For material: Silicone – Extractable components	PASS	
5.	For material: Silicone / Rubber / Plastic – Remaining Peroxide	PASS	
6.	For material: Silicone / Rubber – Volatile Organic Matters	PASS	
7.	For material: Polystyrene and styrene copolymers / Acrylic – Volatile Organic Components	PASS	
8.	For material: Acrylonitrile copolymers – Specific Migration of Acrylonitrile and 1,3-Butadiene for compliance with regulation (EU) No. 10/2011	PASS	
9.	For material: Polyethylene (PE) – Total Chromium, Vanadium and Zirconium content	PASS	
10.	For material: Polypropylene (PP) – Total Chromium, Vanadium, Zirconium and Hafnium content	PASS	
11.	For material: Metal and Metal alloy – Specific Migration of 21 Heavy Metals according to European Directorate for the Quality of Medicines & Healthcare Technical guide PA/PH/EMB (13) 9 and Resolution CM/Res(2013)9	PASS	
12.	For material: Glass and ceramics – Leachable Lead and Cadmium content	PASS	

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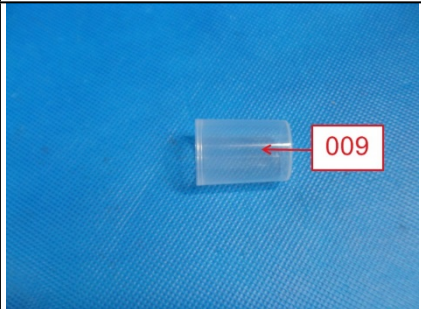
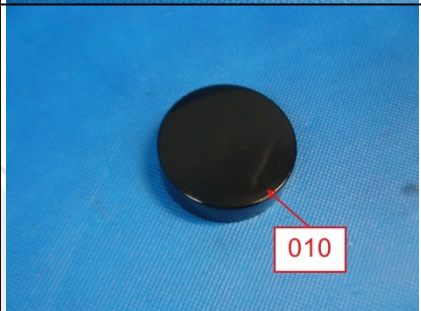
No.	Test Requested	Conclusion	Remarks
13.	For material: Glass and ceramics – Leachable Cobalt content	PASS	
14.	Sensory test – With reference to DIN 10955	PASS	



1. TESTED SUBJECT DESCRIPTION

Sample Number	Item Name	Tested Material Description	Photo
001	Cap(PP)	Black plastic	
002	Silicon	Black silicon	
003	Suckle (PE)	Transparent plastic	
004	Stainless steel	Silvery metal	
005	Suckle (Glass)	Transparent glass	
006	Seal ring (PE)	White plastic	
007	Glass	Brown glass	
008	Glass	Brown glass	

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Sample Number	Item Name	Tested Material Description	Photo
009	PP	Transparent plastic	
010	ABS	Black plastic	



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2. TEST RESULT

2.1. OVERALL MIGRATION TEST FOR PLASTIC

Test method: As specified in Regulation (EU) No. 10/2011; with reference to EN 1186:part 1, part 2, part 3, part 8, part 9 & part 14 :2002.

Simulant Used	Test Condition	Result [mg/dm ²]			Maximum Permissible Limit [mg/dm ²]
		Sample 001	Sample 003	Sample 006	
10% Ethanol	40 °C for 240 hours	5.0	3.1	5.2	10
Rectified Olive Oil	40 °C for 240 hours	< 3.0	< 3.0	< 3.0	10

Simulant Used	Test Condition	Result [mg/dm ²]		Maximum Permissible Limit [mg/dm ²]
		Sample 009	Sample 010	
10% Ethanol	40 °C for 240 hours	3.8	3.5	10
Rectified Olive Oil	40 °C for 240 hours	< 3.0	< 3.0	10

Note:

- “°C” denotes degree Celsius
- “mg/dm²” denotes milligram per square decimeter
- “<” denotes less than
- The specification was quoted from regulation (EU) 10/2011.

2.2. SPECIFIC MIGRATION OF HEAVY METALS (Ba, Co, Cu, Fe, Li, Mn, Zn, Al, Ni) TEST

Test method: As specified in Regulation (EU) No. 10/2011, the sample(s) were migrated with food simulant, followed by Inductively Coupled Plasma Optical Emission Spectrometry (ICP-OES) analysis.

Testing condition and simulant: Distilled water at 40 °C for 72 hour(s).

Test Item	Result [mg/kg]		Maximum Permissible Limit [mg/kg]
	Sample 001	Sample 006	
Barium (Ba)	<0.10	<0.10	1
Cobalt (Co)	<0.05	<0.05	0.05
Copper (Cu)	<0.50	<0.50	5
Iron (Fe)	<1.00	<1.00	48
Lithium (Li)	<0.10	<0.10	0.6
Manganese (Mn)	<0.05	<0.05	0.6
Zinc (Zn)	<1.00	<1.00	5
Aluminium (Al)	<0.05	<0.05	1
Nickel (Ni)	<0.02	<0.02	0.02

Test Item	Result [mg/kg]		Maximum Permissible Limit [mg/kg]
	Sample 009	Sample 010	
Barium (Ba)	<0.10	<0.10	1
Cobalt (Co)	<0.05	<0.05	0.05
Copper (Cu)	<0.50	<0.50	5
Iron (Fe)	<1.00	<1.00	48
Lithium (Li)	<0.10	<0.10	0.6
Manganese (Mn)	<0.05	<0.05	0.6
Zinc (Zn)	<1.00	<1.00	5
Aluminium (Al)	<0.05	<0.05	1
Nickel (Ni)	<0.02	<0.02	0.02

Note:

- “°C” denotes degree Celsius
- “mg/kg” denotes milligram per kilogram foodstuff
- “<” denotes less than
- The specification was quoted from regulation (EU) 10/2011 and it's amendments regulation (EU) 2016/1416, (EU) 2017/752.

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Testing condition and simulant: Distilled water at 40 °C for 240 hour(s).

Test Item	Result [mg/kg]	Maximum Permissible Limit [mg/kg]
	Sample 003	
Barium (Ba)	<0.10	1
Cobalt (Co)	<0.05	0.05
Copper (Cu)	<0.50	5
Iron (Fe)	<1.00	48
Lithium (Li)	<0.10	0.6
Manganese (Mn)	<0.05	0.6
Zinc (Zn)	<1.00	5
Aluminium (Al)	<0.05	1
Nickel (Ni)	<0.02	0.02

Note:

- “°C” denotes degree Celsius
- “mg/kg” denotes milligram per kilogram foodstuff
- “<” denotes less than
- The specification was quoted from regulation (EU) 10/2011 and it's amendments regulation (EU) 2016/1416, (EU) 2017/752.

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2.3. SPECIFIC MIGRATION OF PRIMARY AROMATIC AMINE TEST

Test method: With reference to EN 1186-1: 2002, followed by Kunststoffe im Lebensmittelverkehr, Book 2, Teil B II, XXI. [Detection limit: 0.01 mg/kg]

Testing condition and simulant: Distilled water at 40 °C for 72 hour(s).

Test Item	Result [mg/kg]		Maximum Permissible Limit [mg/kg]
	Sample 001	Sample 010	
Migration of Primary Aromatic Amine	< 0.01	< 0.01	Not Detected (< 0.01 mg/kg)

Note:

- “°C” denotes degree Celsius
- “mg/kg” denotes milligram per kilogram foodstuff
- “<” denotes less than
- The specification was quoted from regulation (EU) 10/2011.

2.4. EXTRACTABLE COMPONENTS TEST

Test method: With reference to Kunststoffe im Lebensmittelverkehr, Book 2, Teil B II, XV.

Simulant Used	Test Condition	Result [%]	Maximum Permissible Limit [%]
		Sample 002	
Distilled water	Reflux for 5 hours	< 0.10	0.5
3% Acetic Acid	Reflux for 5 hours	< 0.10	0.5
10% Ethanol	Reflux for 5 hours	< 0.10	0.5

Note :

- “%” denotes percentage by weight
- “<” denotes less than
- The specification was quoted from Recommendation of the BfR “Kunststoffe im Lebensmittelverkehr” Part XV “Silicone”

2.5. REMAINING PEROXIDE TEST

Test method : With reference to 58th Communication on the testing of plastics, Bundesgesundheitsbl. 40 (1997) 412.

Test Item	Result	Maximum Permissible Limit
	Sample 002	
Remaining Peroxide	Absent	Absent

Note:

- The specification was quoted from Recommendation of the BfR “Kunststoffe im Lebensmittelverkehr Part XV and Part VI.

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2.6. VOLATILE ORGANIC MATTER TEST

Test Method: With reference to 61st Communication on testing of plastics in Bundesgesundheitsbl 46 (2003) 362.

Test Item	Test Condition	Result [%]	Maximum Permissible Limit [%]
		Sample 002	
Volatile Organic Matter	40 °C for 72 hours	< 0.10	0.5

Note:

- “°C” denotes degree Celsius
- “%” denotes percentage by weight
- “<” denotes less than
- The specification was quoted from Recommendation of the BfR “Kunststoffe im Lebensmittelverkehr Part XV “Silicones”

2.7. VOLATILE ORGANIC COMPONENTS TEST

Test Method: With reference to 48th Communication on the testing of plastics, Bundesgesundheitsblatt 25 (1982) 334.

Test Item	Test Condition	Result [mg/dm ²]	Maximum Permissible Limit [mg/dm ²]
		Sample 010	
Volatile Organic Components	90 °C for 24 hours	6.4	15

Note:

- “°C” denotes degree Celsius
- “mg/dm²” denotes milligram per square decimeter
- “<” denotes less than
- The specification was quoted from Recommendation of the BfR “Kunststoffe im Lebensmittelverkehr Part VI “Styrene Copolymers and Graft Polymers, and Mixtures of Polystyrene with other Polymers”

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2.8. SPECIFIC MIGRATION OF ACRYLONITRILE AND BUTADIENE TEST

Test method: As specified in Regulation (EU) No. 10/2011, the sample(s) were migrated with food simulant, followed by Gas Chromatography/Mass Spectrometry (GC-MS) analysis.

Testing condition and simulant: Distilled water at 40 °C for 72 hour(s).

Test Item	Result [mg/kg]	Maximum Permissible Limit [mg/kg]
	Sample 010	
Migration of Acrylonitrile	< 0.01	Not Detected (< 0.01mg/kg)
Migration of 1,3-butadiene	< 0.01	Not Detected (< 0.01mg/kg)

Note:

- “°C” denotes degree Celsius
- “mg/kg” denotes milligram per kilogram foodstuff
- “<” denotes less than
- The specification was quoted from Regulation (EU) No. 10/2011

2.9. TOTAL CHROMIUM, VANADIUM AND ZIRCONIUM CONTENT TEST

Test method: Microwave digestion, followed by Inductively Coupled Plasma Optical Emission Spectrometry (ICP-OES) analysis.

Test Item	Result [mg/kg]		Maximum Permissible Limit [mg/kg]
	Sample 003	Sample 006	
Chromium (Cr)	< 10	< 10	10
Vanadium (V)	< 15	< 15	20
Zirconium (Zr)	< 15	< 15	100

Test Item	Result [mg/kg]	Maximum Permissible Limit [mg/kg]
	Sample 009	
Chromium (Cr)	< 10	10
Vanadium (V)	< 15	20
Zirconium (Zr)	< 15	100

Note:

- “mg/kg” denotes milligram per kilogram
- “<” denotes less than
- The specification was quoted from Recommendation of the BfR “Kunststoffe im Lebensmittelverkehr” Part III “Polyethylene”.

2.10. TOTAL CHROMIUM, VANADIUM, ZIRCONIUM AND HAFNIUM CONTENT TEST

Test method: Microwave digestion, followed by Inductively Coupled Plasma Optical Emission Spectrometry (ICP-OES) analysis.

Test Item	Result [mg/kg]	Maximum Permissible Limit [mg/kg]
	Sample 001	
Chromium (Cr)	< 10	10
Vanadium (V)	< 15	20
Zirconium (Zr)	< 15	100
Hafnium (Hf)	< 15	100

Note:

- “mg/kg” denotes miligram per kilogram
- “<” denotes less than
- The specification was quoted from Recommendation of the BfR “Kunststoffe im Lebensmittelverkehr” Part VII “Polypropylene”.



2.11. SPECIFIC MIGRATION OF HEAVY METAL CONTENT TEST

Test method: The sample(s) were extracted with food simulant , followed by analysis using Inductively Coupled Plasma Optical Emission Spectrometry (ICP-OES) and Inductively Coupled Plasma Mass Spectrometry(ICP-MS).

Testing condition and simulant: artificial tap water at 40 °C for 72 hour(s).

No.	Test Item		Result [mg/kg] (3 rd migration)	Limit for 3 rd migration [mg/kg]
			Sample 004	
1.	Barium	(Ba)	<0.1	1.2
2.	Copper	(Cu)	<0.1	4
3.	Iron	(Fe)	<0.1	40
4.	Tin	(Sn)	<0.5	100
5.	Chromium	(Cr)	<0.05	0.250
6.	Manganese	(Mn)	<0.1	1.8
7.	Zinc	(Zn)	< 0.1	5
8.	Aluminum	(Al)	<0.1	5
9.	Lithium	(Li)	<0.005	0.048
10.	Beryllium	(Be)	<0.002	0.01
11.	Vanadium	(V)	< 0.002	0.01
12.	Nickel	(Ni)	<0.05	0.14
13.	Cobalt	(Co)	<0.002	0.02
14.	Arsenic	(As)	<0.0004	0.002
15.	Molybdenum	(Mo)	<0.002	0.12
16.	Silver	(Ag)	<0.002	0.08
17.	Cadmium	(Cd)	<0.0004	0.005
18.	Antimony	(Sb)	<0.005	0.04
19.	Mercury	(Hg)	< 0.0005	0.003
20.	Thallium	(Tl)	< 0.0001	0.0001
21.	Lead	(Pb)	< 0.01	0.010

Note:

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- “<” denotes less than

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Testing condition and simulant: artificial tap water at 40 °C for 72 hour(s).

No.	Test Item		Result [mg/kg] (1 st + 2 nd Migration)	Limit for 1 st +2 nd migration [mg/kg]
			Sample 004	
1.	Barium	(Ba)	<0.2	8.4
2.	Copper	(Cu)	<0.2	28
3.	Iron	(Fe)	<0.2	280
4.	Tin	(Sn)	<1.0	700
5.	Chromium	(Cr)	<0.1	1.75
6.	Manganese	(Mn)	<0.2	12.6
7.	Zinc	(Zn)	< 0.2	35
8.	Aluminum	(Al)	<0.2	35
9.	Lithium	(Li)	<0.01	0.336
10.	Beryllium	(Be)	< 0.004	0.07
11.	Vanadium	(V)	< 0.004	0.07
12.	Nickel	(Ni)	<0.1	0.98
13.	Cobalt	(Co)	<0.004	0.14
14.	Arsenic	(As)	<0.0008	0.014
15.	Molybdenum	(Mo)	<0.004	0.84
16.	Silver	(Ag)	<0.004	0.56
17.	Cadmium	(Cd)	<0.0008	0.035
18.	Antimony	(Sb)	<0.01	0.28
19.	Mercury	(Hg)	< 0.001	0.021
20.	Thallium	(Tl)	< 0.0002	0.0007
21.	Lead	(Pb)	< 0.02	0.07

Note:

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- “mg/kg” denotes milligram per kilogram foodstuff
- “<” denotes less than

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2.12. LEACHABLE LEAD AND CADMIUM CONTENT TEST

Test method: With reference to BS EN 1388: Part 1: 1996 and BS EN 1388: Part 2: 1996.

Test Item	Unit	Result	Maximum Permissible Limit
		Sample 005	
Leachable Lead	mg/dm ²	< 0.10	0.8
Leachable Cadmium	mg/dm ²	< 0.02	0.07

Note:

- “mg/dm²” denotes milligram per square decimeter
- “<” denotes less than
- The specification was quoted from directive 84/500/EEC for article as flatware (category 1).

Test Item	Unit	Result		Maximum Permissible Limit
		Sample 007	Sample 008	
Leachable Lead	mg/L	< 0.10	< 0.10	4.0
Leachable Cadmium	mg/L	< 0.05	< 0.05	0.3

Note:

- “mg/L” denotes milligram per Litre
- “<” denotes less than
- The specification was quoted from directive 84/500/EEC for article as hollowware (category 2).

2.13. LEACHABLE COBALT CONTENT TEST

Test method: With reference to BS EN 1388: Part 1: 1996 and BS EN 1388: Part 2: 1996.

Test Item	Unit	Result			Maximum Permissible Limit
		Sample 005	Sample 007	Sample 008	
Leachable Cobalt	mg/L	<0.05	<0.05	<0.05	0.1

Note:

- “mg/L” denotes milligram per Litre
- “<” denotes less than
- The specification was quoted from Germany Bavarian State Ministry of Justice and Consumer Protection.

2.14. SENSORY TEST

Test method: With reference to DIN 10955: 2004. The submitted sample was treated with food stimulant. After this treatment, treated water was examined by panels with regard to any divergence in smell and taste.

Testing condition and simulant: Distilled water at 40 °C for 240 hour(s).

Test Item	Grading Result		Recommended Level
	Sample 001+002+003+ 006+009	Sample 004	
Transfer of Smell	1	1	< 2.5
Transfer of Taste	0	0	< 2.5

Test Item	Grading Result	Recommended Level
	Sample 005+007+008	
Transfer of Smell	1	< 2.5
Transfer of Taste	0	< 2.5

Note:

- “<” denotes less than
- Explanation for grading are listed as below:
 Grading 0: No perceptible taste/smell deviation
 Grading 1: Just perceptible taste/smell deviation
 Grading 2: Weak taste/smell deviation
 Grading 3: Clear taste/smell deviation
 Grading 4: Strong taste/smell deviation
